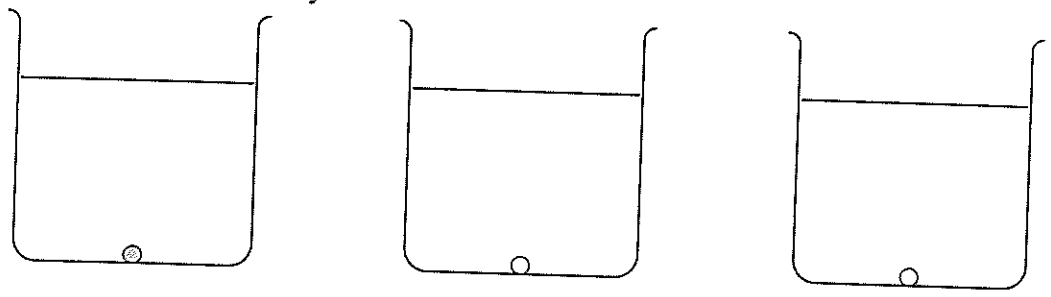
Diffusion

Name _____ **Class** _____

1 Jenna put a purple crystal into a beaker of water at 10 o'clock. She came back to look at the beaker at 12 o'clock, and again at 4 o'clock.

Colour in the beakers to show how far the purple colour would have spread at each time. The first one has been done for you.



10 o'clock *12 o'clock* *4 o'clock*

2 Fill in the gaps in these sentences, using words from the box. You may need to use some words more than once.

The colour in Jenna's beaker spread out by _____ . The water _____ are moving around all the time. They bump into _____ from the purple crystal, and _____ them around. The purple gradually _____ out through the _____ .

diffusion	**move**	**particles**	**spreads**	**water**

3 Mr Collins has got some new aftershave. Danny can smell it as soon as Mr Collins comes into the lab. Ramesh does not smell it until two minutes later.

a Who is sitting nearest to the lab door? _____

b Explain your answer _____

_____

Sublimation

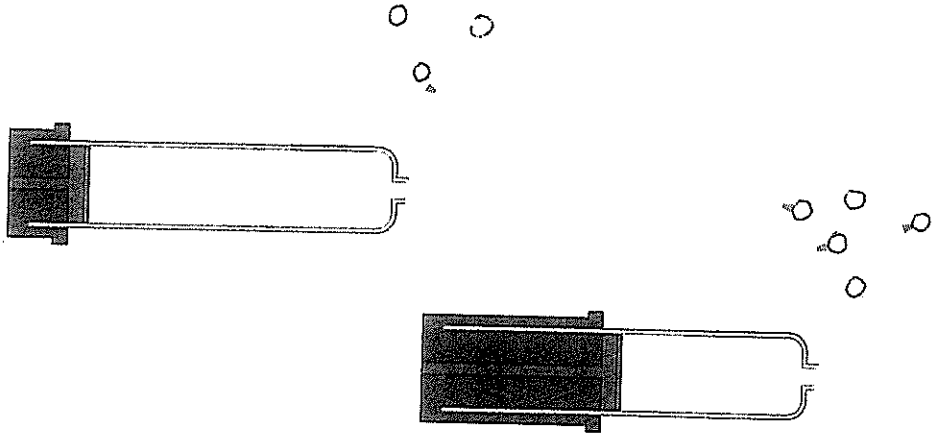
Some substances, like iodine and graphite, do not melt when heated but change straight from a solid into a gas. This change is reversible like melting and evaporation. These unusual changes of state, solid to gas and gas to solid are both called **sublimation**.



Figure 5.4 *Sublimation of iodine.*

Compressing matter

It is hard to compress (squash) solids and liquids as the particles are so close together they have little space between them. Gases, however, are fairly easy to compress. The particles are far apart in gases and they can be squeezed into a smaller space by forcing the particles closer together. Figure 5.5 shows what happens to the particles in air when they are compressed in a syringe.



igure 5.5 *Compressing gases.*

Diffusion

Diffusion occurs when the particles of one substance spread out nd mix with the particles of another substance. Diffusion occurs nost rapidly in gases as the particles in gases move fastest and are urthest apart. If two gas jars are placed together and the lid removed, he gas particles rapidly mix together due to the movement of their articles.

is due to diffusion of gas particles that we are able to smell ıbstances like perfume or food. This happens as the gas particles of ıe smelly substance are moving about in all directions and eventually



As particles slow down, the temperature of a substance falls. Eventually the particles must stop moving and it cannot get any colder. This is the minimum temperature possible, it occurs at –273.16°C and is called absolute zero.

18

Boiling point of water

Does the boiling point of water change if salt is added?

Apparatus

- **Beaker**
- **Water**
- **Thermometer**
- **Salt**
- **Spatula**
- **Balance**

Prediction

1 Do you think the salt will have an effect on the boiling point? Will it make it higher or lower? Why do you think this?

Planning

2 Write a method for your investigation. You need to think about these things:

- How will you measure the boiling point?
- How will you make it a fair test?
- How many experiments will you do?
- How much salt will you add?
- How much water will you use?
- How will you present your results?

Show your plan to your teacher, then carry out your investigation.

Recording your results

3 Put your results in a neat table.

Considering your results/conclusion

4 What have you found out? Was this what you predicted?

Evaluation

5 If you had time to do your investigation again, could you improve it? Explain your answer.

[planning, observing, presenting, considering, evaluating]

Theories

1
In ice there is a regular arrangement of particles. The particles build up in hexagon rings. This explains the shape of snowflakes. There are gaps in the middle of the rings, which means that ice is less dense than water.

2
Hot water takes up more space than the same amount of cold water. In any volume, there will be fewer particles of hot water than cold.

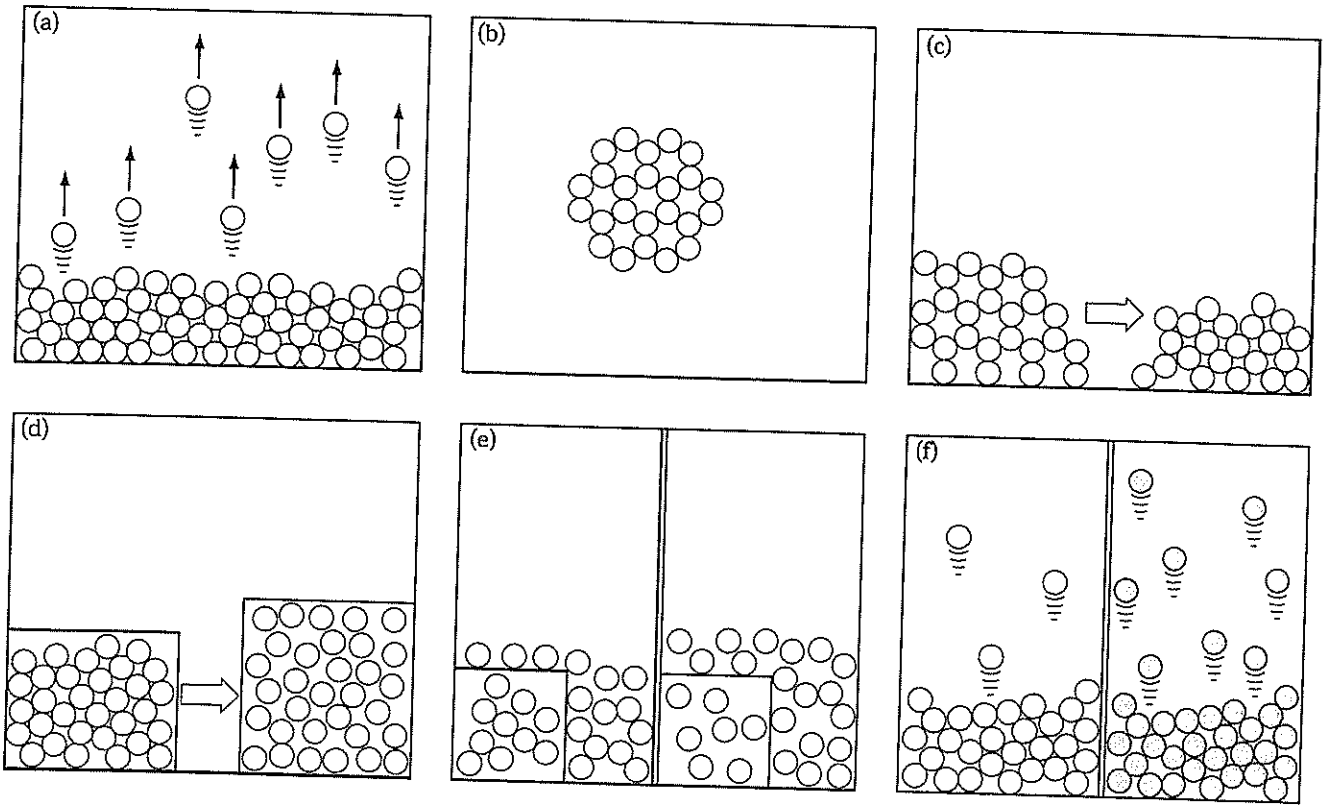
3
The bonds between particles of petrol are weaker than the bonds between particles of water. It is easier for the petrol particles to escape from the liquid and evaporate into the air.

4
When the water boils, the heat gives the particles enough energy to escape from the liquid.

5
As the water heats up, the particles gain energy and move faster. This movement means that the water takes up more space.

6
When the ice melts, the hexagon pattern is broken up. The particles do not have any particular arrangement, and there are fewer gaps between them.

Particle models



Using ideas about particles

Scientists can use their imagination to make **models** of the way they think that particles are arranged and the way that they move. These models can explain the **observations** that we make, and can sometimes be used to make further **predictions**.

1 Cut out all the boxes.

Think about which of the 'theory' boxes best matches each observation.

Then look at the 'particle model' boxes and find the correct diagram.

When you have matched up the six sets correctly, stick them into your book.

2 Use the information in the boxes to make the following predictions. In each case, explain your reasoning. If you cannot make a prediction, try to explain why not.

a What happens to the size of a solid if you heat it?

b What happens to the size of a solid if you cool it down?

c Which evaporates the fastest, petrol or perfume?

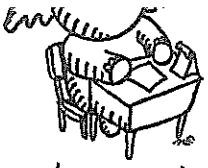
d What will happen to the temperature of the ice while it is melting?

[knowledge]

Observations

A When water boils, it stays at the same temperature as long as there is some water left.	B Ice is a solid which floats on water. Snowflakes are always six-sided shapes.	C If you spill some petrol, it evaporates very quickly. Water takes longer to evaporate.
D When the water boils, the heat gives the particles enough energy to escape from the liquid.	E When ice melts, it takes up less space.	F When you heat water, it expands (gets bigger).

DO NOT MAKE A DOUBLE SIDED PHOTOCOPY OF THIS WORKSHEET

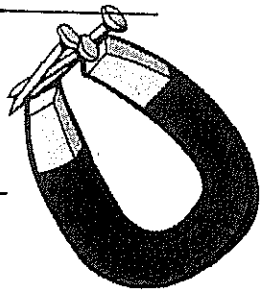


The Particle Theory

Read page 224 and summarise the first two paragraphs in your own words in the space below.

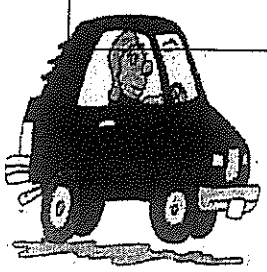
What are the five points of the particle theory of matter?

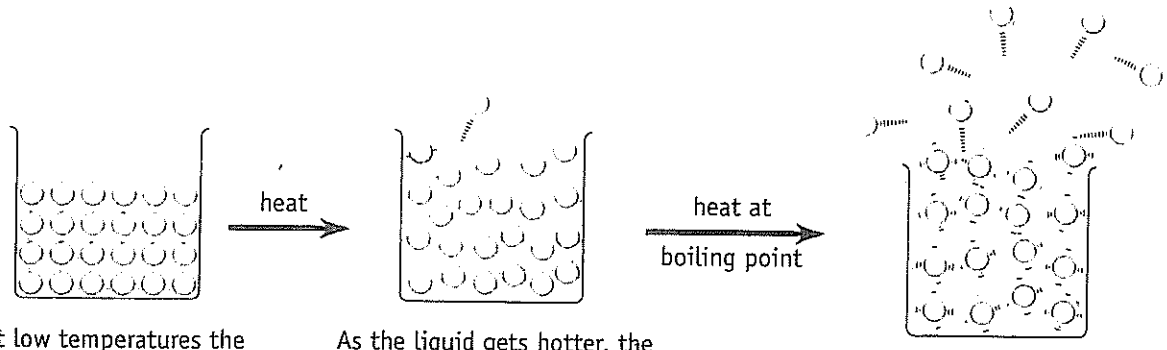
1.
2.
3.
4.
5.



Solids

In this box, draw a diagram of how the particles are arranged in a solid	Explain your diagram





At low temperatures the particles in the liquid move slowly.

As the liquid gets hotter, the particles move more quickly. Some have enough energy to escape (evaporate).

At the boiling point all the particles have enough energy to evaporate.

Fig 24 Using the particle theory to explain evaporation and boiling of a liquid.

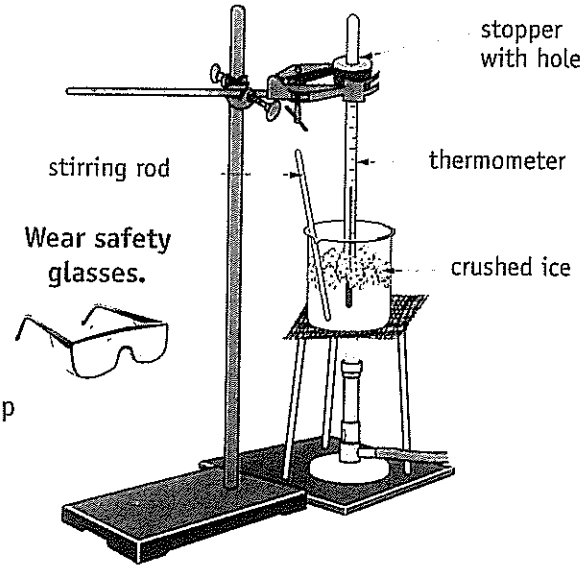
EXPERIMENT 3 MELTING AND BOILING

Aim

To measure and graph the temperature as ice melts to water and then boils.

Materials

- small beaker, eg 250 mL
- crushed ice
- thermometer (−10 to 110°C)
- burner, tripod, gauze and heatproof mat
- stopwatch
- stirring rod
- metal stand and clamp
- stopper with hole to hold thermometer in clamp
- graph paper



PLANNING AND SAFETY CHECK

Read through the Method section.

- ✦ Which is the independent variable and which is the dependent variable? How do you know which is which?
- ✦ Which safety precautions will be necessary?

Design a data table, with columns for the two variables to be measured.

Method

1 Set up the apparatus as shown.

2 Half-fill the beaker with crushed ice, and measure its temperature. (Remember to wait until the reading is steady.)
✍ Record the temperature of the ice in your data table.

3 Light the burner and start timing.

4 Measure the temperature every minute. Use the stirring rod to stir gently before each

CHAPTER 11

ACTIVITY

1 Make a model for matter by putting some ball bearings in a flat dish or box.

What do the ball bearings represent? What does the dish or box represent?

Draw the arrangement of the ball bearings.

What state of matter does this represent?

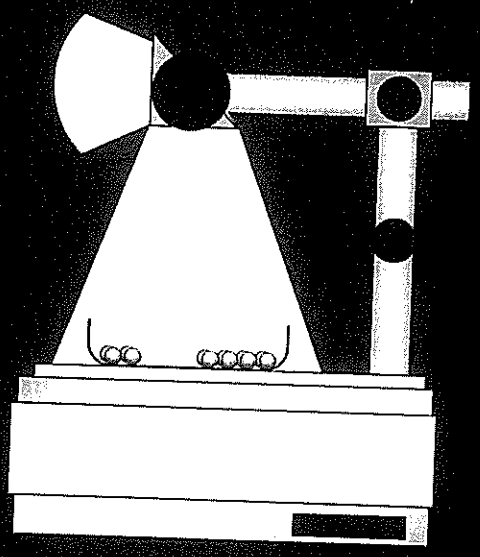
2 Shake the dish gently so that the ball bearings move about.

Describe the new arrangement of ball bearings.

What state of matter does this represent?

3 Shake the dish vigorously.

Describe the new arrangement. What state of matter does it represent?



Your teacher may demonstrate this model using a glass dish on an overhead projector.

Explaining changes of state

We can use the particle theory to explain changes of state. When a solid is heated, its particles gain more energy and vibrate more. This makes the solid expand—get bigger. At the melting point the particles vibrate so much that they break away from their positions. When this happens the solid becomes a liquid.

When a liquid is heated, its particles have more energy and move faster. They bump into each other more energetically and bounce further apart. This makes the liquid expand. At the boiling point, the particles have enough energy to overcome the forces holding them together. They break away from the liquid and form a gas. (See Fig 24 on the opposite page.)

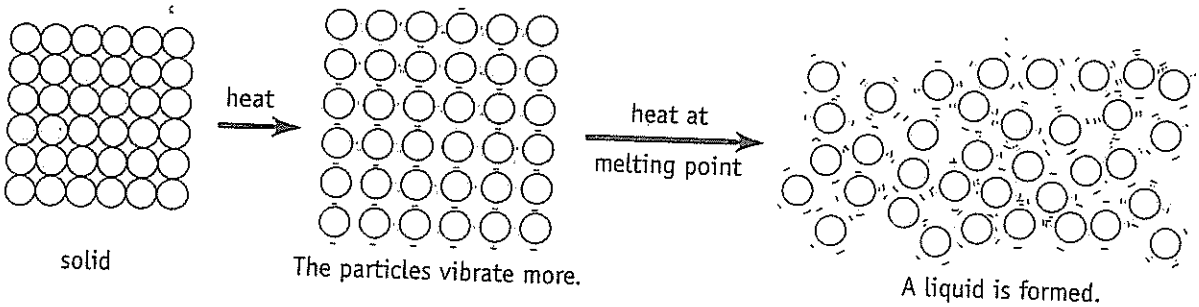


Fig 23 Using the particle theory to explain melting.

Summing up states

Name _____ **Class** _____

There are three states of matter: _____ , _____ , and _____ .

	Solids	**Liquids**	**Gases**
Shape	Have a _____ shape	Have no _____ _____	Have no fixed _____
Volume	Have a _____ volume	Have a _____ volume	Have no _____ _____
Density	Have a _____ density	Are usually _____ dense than solids	Have a very low _____
Ability to flow	Do _____ flow	_____ quite easily	_____ very easily
Ability to be squashed	_____ be squashed	_____ to squash	Quite easy to _____
Closeness of particles	Particles are very _____ together. They can hardly _____ at all	Particles are quite t _____ together. They can _____ past each other	Particles are _____ out and free to _____ around
Use these spaces to draw diagrams to show how the particles are arranged			
Examples		water	

CHAPTER 11

The particle theory

More than 2000 years ago in ancient Greece a thinker called Democritus made this hypothesis: *all matter, living and non-living, is made of tiny particles too small to be seen*. His idea was that if you kept cutting something into smaller and smaller pieces you would eventually come to the smallest possible particles—the building blocks of matter. He used the word *atomos* (which in Greek means 'cannot be divided') to describe these tiniest particles. This is where the word 'atom' comes from.

Since then scientists have done many tests with matter, and the results have always agreed with Democritus' hypothesis. Such a hypothesis that is supported by many experimental results is called a **theory**. So the hypothesis that matter is made up of tiny particles too small to see is now called the **particle theory** of matter.

This particle theory can be used to explain the properties of solids, liquids and gases.

The particle theory of matter

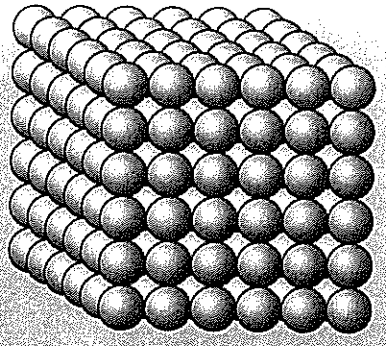
1 All matter is made up of tiny particles too small to see.

2 There are spaces between the particles.

3 There are attractive forces between particles. The weaker these forces are, the further apart the particles are.

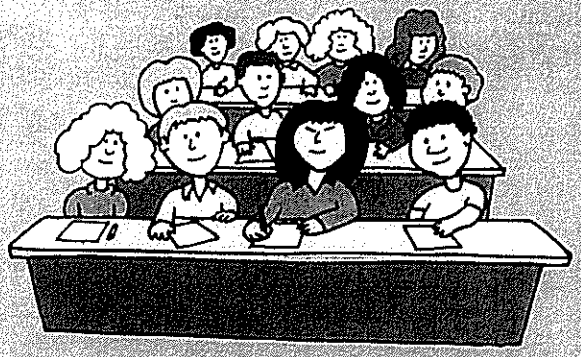
4 The particles are always moving.

5 At high temperatures the particles move faster than they do at low temperatures.

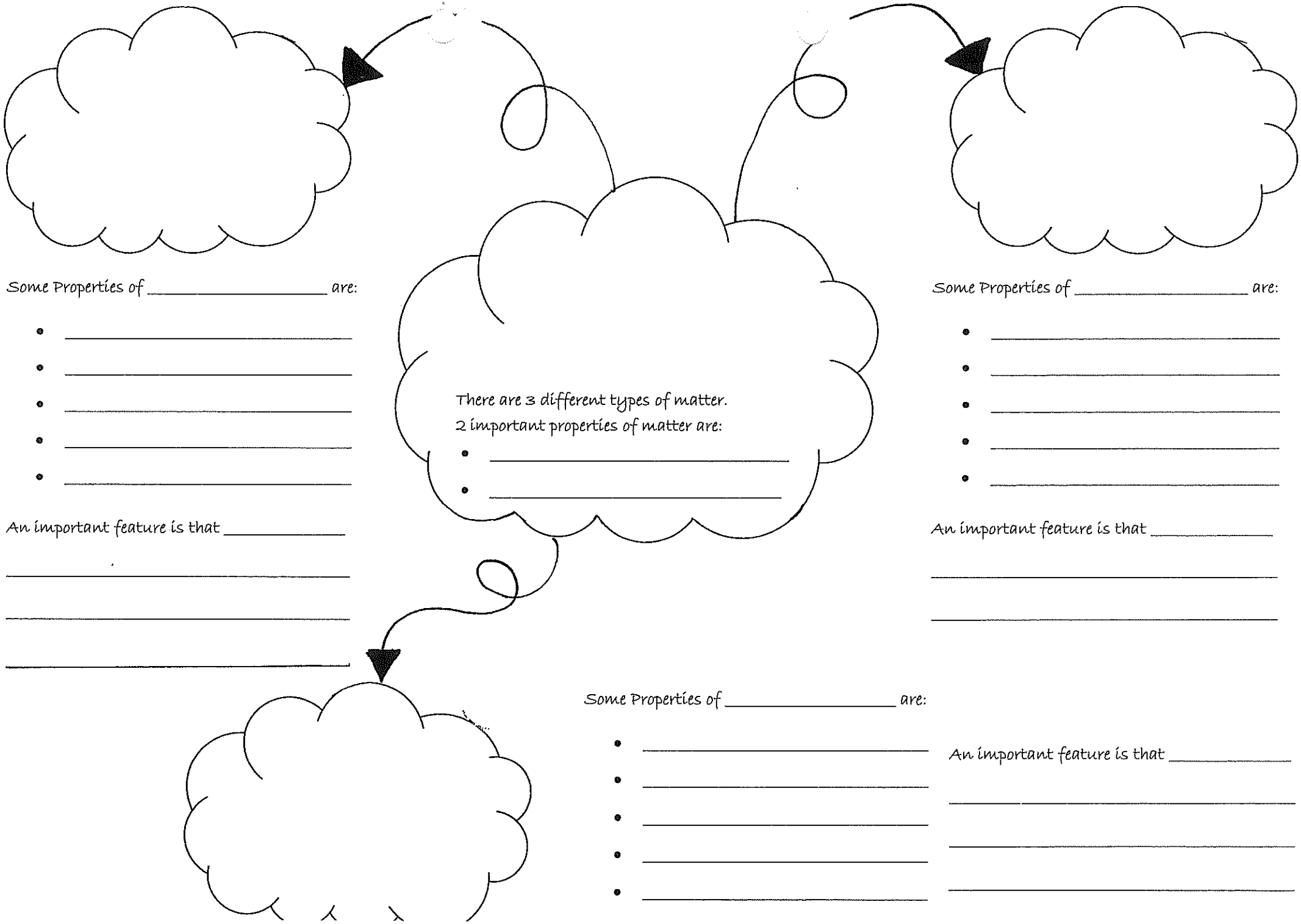


The particles in a solid (eg steel) are packed tightly in a fixed pattern. There are strong forces holding them together, so they cannot leave their positions. The only movements they make are tiny vibrations to and fro.

We cannot see these invisible particles, but we can use a **model**. For example, we can represent the particles by the students in your class. When everyone is sitting down, the class is a model for a solid.



The word model has a special meaning in science. It is not the latest *model* car or a fashion *model*. It is a way of representing something that is too small to be seen, or too large or complicated to be studied easily. A model is not the real thing. It is only a representation which helps you understand or explain something.



There are 3 different types of matter.
2 important properties of matter are:

Some Properties of _____ are:

Some Properties of _____ are:

An important feature is that _____

Some Properties of _____ are:

An important feature is that _____

Properties cards

Solids	Liquids
Gases	

Difficult to squash	Flow easily	Do not flow
Easy to squash	Fixed shape	Fixed volume
High density	Flow easily	No fixed volume
Fixed volume	Low density	No fixed shape
No fixed shape	Difficult to squash	Quite high density

[knowledge]